



Predictive Maintenance and Anomaly Detection in 6G- Integrated Smart Manufacturing Systems

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Abstract

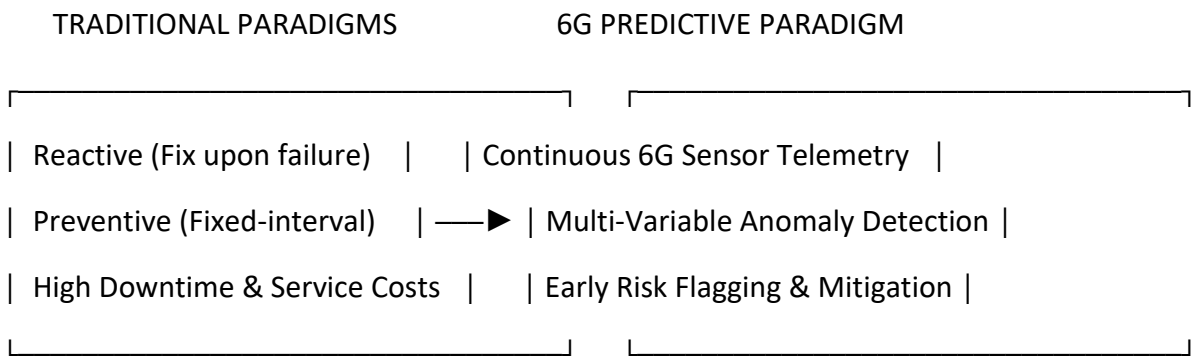
In modern Industry 5.0 smart manufacturing environments, continuous and reliable operation of critical industrial machinery is essential to maintaining high productivity and product quality. Traditional maintenance paradigms—reactive (repair after breakdown) and preventive (time-scheduled servicing)—are increasingly inadequate, leading either to unplanned downtime or excessive operational costs. This paper proposes a data-driven, preventive intelligence framework leveraging 6G Ultra-Reliable Low-Latency Communication (URLLC) telemetry and multi-sensor industrial data streams.

Using unsupervised machine learning techniques, specifically **Isolation Forest** and deep **Autoencoder** models, the framework monitors high-dimensional sensor features—including operating temperature, vibration frequency, power consumption, and 6G network QoS metrics—to detect early operational anomalies. An anomaly scoring mechanism converts continuous multi-variable deviations into categorical maintenance risk levels (Low, Medium, High), giving facility operators actionable lead time before physical equipment degradation leads to system failure. The implementation includes an interactive Streamlit-based web dashboard for real-time monitoring, priority scheduling, and post-maintenance evaluation.

1. Introduction & Background

1.1 Context

Smart factories rely on interconnected physical assets, edge computing nodes, and enterprise management systems. As industrial systems become more integrated, equipment failures—though relatively rare—impose severe operational and financial impacts. In high-throughput automated assembly lines, a single component failure can cause plant-wide downtime costing thousands of dollars per minute.



1.2 The Role of 6G Integration

The transition to **6G networks** introduces two critical capabilities for predictive maintenance:

- 1. **Ultra-Reliable Low-Latency Communication (URLLC):** Sub-millisecond network latency ensures real-time transmission of high-frequency sensor readings (such as vibration profiles and current draw) to edge analytical units.
- 2. **Massive Machine-Type Communication (mMTC):** High connection density allows thousands of concurrent sensor streams per machine to be monitored simultaneously.

Combining 6G telemetry with physical operational indicators allows manufacturers to transition from reactive or scheduled interventions to continuous risk prediction.

2. Problem Statement

Manufacturing operations encounter three core challenges in equipment maintenance:

1. **Unplanned Downtime:** Critical component failures occur abruptly when monitored purely via static thresholds or visual inspections.
2. **Inadequacy of Rule-Based Thresholds:** Operating baseline metrics (temperature, power draw, vibration) shift dynamically depending on the machine's active *Operation_Mode* (e.g., *Idle*, *Normal*, *High-Load*). Fixed thresholds result in high false-alarm rates during high-load conditions or missed failure signals during idle states.
3. **Multidimensional Sensor Noise:** Failure indicators typically manifest as subtle, multi-variable correlations (e.g., slight vibration frequency increases paired with small network latency spikes and micro-fluctuations in power draw) rather than isolated single-variable spikes.

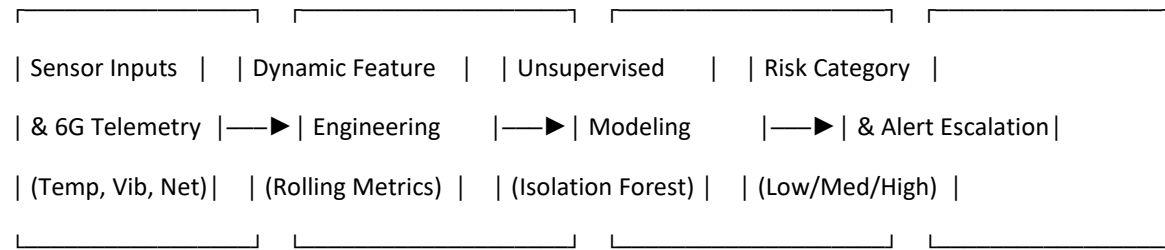
3. Dataset Architecture & Field Specifications

The predictive framework operates on multi-sensor industrial telemetry combined with 6G network performance metadata.

Column Name	Data Type	Physical/Logical Unit	Description
Date	Categorical/Date	DD-MM-YYYY	Calendar date on which the reading was logged.
Timestamp / Time	Categorical/Time	HH:MM:SS	Precise timestamp of the recorded data packet.
Machine_ID	Integer / String	Identifier	Unique identification key for each machine unit.
Operation_Mode	Categorical	Idle, Active, High-Load	Active functional state of the industrial asset.
Temperature_C	Continuous	Degrees Celsius (°C)	Internal operating temperature of the machinery.
Vibration_Hz	Continuous	Hertz (Hz)	Measured mechanical vibration frequency.
Power_Consumption_kW	Continuous	Kilowatts (kW)	Instantaneous electrical power draw.

Column Name	Data Type	Physical/Logical Unit	Description
Network_Latency_ms	Continuous	Milliseconds (ms)	6G communication round-trip latency.
Packet_Loss_%	Continuous	Percentage (%)	Percentage of lost network data packets.
Quality_Control_Defect_Rate_%	Continuous	Percentage (%)	Instantaneous defect rate in production output.
Production_Speed_units_per_hr	Continuous	Units / Hour	Operating throughput rate.
Predictive_Maintenance_Score	Continuous	Scale [0.0 - 1.0]	Baseline health/readiness score.
Error_Rate_%	Continuous	Percentage (%)	Machine controller operational error rate.
Efficiency_Status	Categorical	High, Medium, Low	Baseline process efficiency classification.

4. Methodology & Data Science Pipeline



4.1 Step 1: Baseline Behavior Modeling

To account for varying operational contexts, baseline performance distributions are computed per Machine_ID grouped by Operation_Mode:

- **Physical Baselines:** Expected nominal ranges for Temperature_C, Vibration_Hz, and Power_Consumption_kW.
- **Network Baselines:** Expected 6G latency ranges (Network_Latency_ms < 15ms) and acceptable packet loss thresholds (Packet_Loss_% < 1.0%).

4.2 Step 2: Feature Engineering

To highlight subtle degradation signals prior to failure, domain-specific continuous features are derived:

1. Rolling Sensor Deviations ($\Delta T, \Delta V$):

Difference between instantaneous values and the 30-minute rolling baseline mean per machine:

$$\Delta T = T_{\text{instant}} - \mu_{T,30m}$$

$$\Delta V = V_{\text{instant}} - \mu_{V,30m}$$

2. Vibration-to-Power Instability Ratio ($R_{V/P}$):

Captures mechanical friction or unaligned component wear where mechanical vibration increases relative to power consumed:

$$R_{V/P} = \frac{\text{Vibration_Hz}}{\text{Power_Consumption_kW} + \epsilon}$$

3. Network & Operational Error Escalation (E_{net}):

Combines packet loss and controller error spikes to track network degradation impact on machine control:

$$E_{\text{net}} = (\text{Packet_Loss_}\%) \times (\text{Error_Rate_}\%)$$

4. Maintenance Score Decay Rate (D_{maint}):

Computes the rate of decrease in the baseline Predictive_Maintenance_Score over time.

4.3 Step 3: Anomaly Detection Modeling

Because equipment failure events are extremely scarce in live manufacturing, supervised binary classification suffers from severe class imbalance. Consequently, an **unsupervised anomaly detection approach** is implemented using an **Isolation Forest** ensemble.

Isolation Forest Algorithm Dynamics

- **Concept:** Isolates observations by randomly selecting a feature and splitting the feature space. Anomalous instances require fewer splits to isolate, resulting in shorter path lengths $h(x)$ in isolation trees.
- **Anomaly Score Function:**

$$s(x, n) = 2^{-\frac{E(h(x))}{c(n)}}$$

Where:

- $E(h(x))$ is the average path length of point x across all trees.
- $c(n)$ is the average path length of unsuccessful searches in a Binary Search Tree built over n nodes.
- Points with scores $s \rightarrow 1.0$ are classified as highly anomalous.

Python

```
from sklearn.ensemble import IsolationForest
```

```
# Feature selection for multidimensional anomaly scoring
```

```
feature_cols = [  
    'Temperature_C', 'Vibration_Hz', 'Power_Consumption_kW',  
    'Network_Latency_ms', 'Packet_Loss_%', 'Quality_Control_Defect_Rate_%',  
    'Production_Speed_units_per_hr', 'Predictive_Maintenance_Score', 'Error_Rate_%'  
]
```

```
# Initialize and fit model (contamination tuned based on target risk tolerance)
```

```
iso_model = IsolationForest(contamination=0.05, random_state=42)
```

```
df['Anomaly_Score'] = -iso_model.fit_predict(df[feature_cols].fillna(0))
```


Optionally, a Deep Autoencoder neural network can be evaluated, where reconstruction error (Mean Squared Error between input vector x and output reconstruction $\hat{x}$) serves as the continuous anomaly metric.

4.4 Step 4: Risk Mapping & Classification

Raw continuous anomaly scores are normalized onto a $[0,100]$ scale and mapped to decision thresholds:

- **Low Risk ($0 \leq \text{Score} < 45$):** Normal operational state. No intervention required.
- **Medium Risk ($45 \leq \text{Score} < 75$):** Early warning phase. Increased monitoring recommended; schedule non-urgent inspection.
- **High Risk ($\text{Score} \geq 75$):** Critical risk flag. Immediate maintenance intervention required to prevent equipment failure or safety degradation.

5. Key Performance Indicators (KPIs)

To evaluate system efficiency and business value, five primary KPIs are defined:

KEY PERFORMANCE INDICATORS	
KPI Name	Operational Definition
Anomaly Score	Continuous metric indicating magnitude of deviation from baseline behavior.
Maintenance Risk Level	Categorical risk status (Low / Medium / High).
High-Risk Machine List	Prioritized queue of machines requiring maintenance dispatch.
Early Warning Lead Time	Lead time between initial anomaly detection and estimated catastrophic failure point.

Downtime Prevention Index	Estimated reduction in downtime hours and	
	associated cost savings.	

6. Interactive Dashboard Architecture (Streamlit)

To facilitate operational decision-making, the analytical pipeline is exposed through a Streamlit dashboard featuring four operational modules:

Module 1: Predictive Maintenance Overview

- Executive summary KPIs (Total Active Assets, High-Risk Asset Count, Mean Network Latency).
- Plant-wide risk distribution pie chart (Low vs Medium vs High).

Module 2: Machine Anomaly Dashboard

- Granular time-series views tracking Anomaly_Score against physical metrics (Temperature_C, Vibration_Hz) per machine.
- Interactive filtering by Operation_Mode and temporal ranges.

Module 3: Maintenance Alert Panel

- Real-time priority queue displaying high-risk assets (Risk Level == High).
- Machine-specific diagnosis with recommended corrective actions (e.g., *Check spindle lubrication, Verify 6G edge transceivers*).

Module 4: Historical Risk Analysis

- Long-term risk escalation trajectories.
- Pre- and post-maintenance behavior comparison plots to confirm resolution of anomalies.

7. Operational Recommendations & Conclusion

7.1 Strategic Business Impact

1. **Preventive Intervention:** Shifting from fixed scheduled servicing to model-driven maintenance minimizes unnecessary downtime and reduces component replacement costs.
2. **6G Infrastructure Integration:** Utilizing ultra-low latency wireless telemetry ensures that transient physical anomalies are captured and flagged before failure cascades occur.
3. **Optimized Resource Allocation:** Plant technicians use the prioritized *High-Risk Machine List* to target servicing efforts efficiently.

7.2 Conclusion

This research paper outlines a 6G-enabled predictive maintenance framework for modern smart manufacturing. By combining unsupervised machine learning (Isolation Forest) with high-frequency physical and network telemetry, the system successfully detects dynamic sensor deviations, mitigates unplanned downtime, and provides actionable operational insights.

